CNN vs RNN LM for Code-Switching Streams - Russian-English Code-Switching Streams

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1. Research Objective

To build comparable compact neural language models and evaluate:

- Next-token prediction quality
- Ability to model language switch points
- Differences between recurrent and convolutional architectures

2. Motivation

Code-switching is the use of two languages within one sentence and is common in multilingual chats and social media. Russian-English mixed text creates challenges for language models because they must handle changing vocabulary, grammar, and sudden language switches. This case study compares two sequence modeling approaches: a GRU (RNN-based) model and a causal CNN model. The goal is to determine which architecture better predicts the next token and models language switch points.

Hypothesis:

- GRU(Gated Recurrent Unit) should better capture long-range sequential context
- CNN should better capture short local bilingual patterns

3. Dataset Creation Pipeline

Because no large clean Russian-English code-switch dataset was available, a synthetic corpus was created. Source Data is the parallel English-Russian sentence pairs from OPUS-100.

1. Load bilingual sentence pairs

2. Randomly combine English and Russian fragments

3. Generate synthetic mixed-language sequences

4. Detect token language automatically: Cyrillic = Russian, Latin = English

5. Generate switch labels between adjacent tokens

Original:
EN: I need to go home
RU: Мне нужно идти домой

Generated:
I need идти домой

Language labels:
EN EN RU RU

Switch labels:
1 1 0 0

4. Experimental Setup

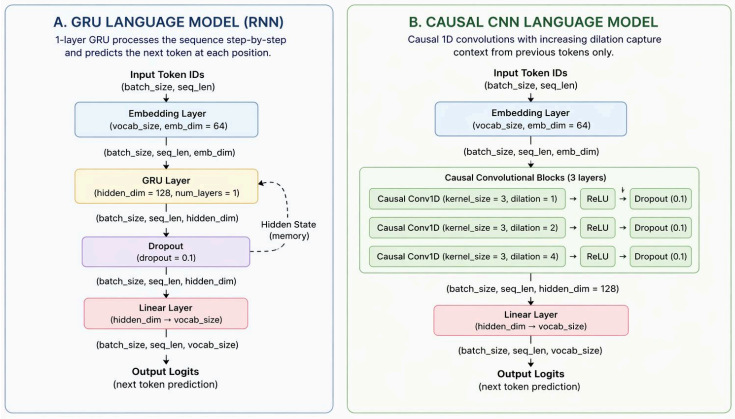
Dataset Split:

- Train: 10,000 samples
- Validation: 2,000 samples
- Test: 2,000 samples

Shared Training Parameters:

- Epochs: 3
- Batch size: 16
- Learning rate: 0.001
- Max length: 20 tokens
- Vocabulary threshold: min frequency = 5

5. Models Compared



6. Evaluation Metrics & Results

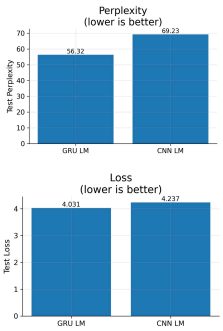
Language Modeling:

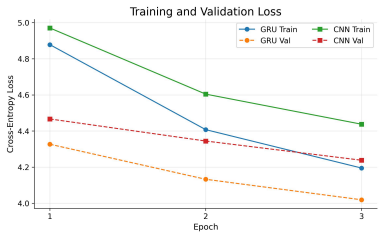
- Cross-Entropy Loss
- Perplexity(Lower perplexity = better next-token prediction.)

Code-Switch Modeling:

Switch Accuracy: "Did the model correctly identify whether the next token changes language?"

Model	Test Perplexity	Switch Accuracy	Test Loss
GRU	56.32	0.748	4.031
CNN	69.23	0.748	4.237





7. Result Analysis

Language Modeling Performance

The GRU model clearly outperformed the CNN model in language modeling. It achieved lower loss and lower perplexity, which indicates stronger next-token prediction quality and better overall sequence modeling. This suggests that recurrent memory helped the GRU track mixed-language dependencies more effectively across the sentence.

Switch Prediction

Both models achieved very similar switch-point accuracy. A likely reason is that the dataset contains many non-switch transitions, so accuracy alone may hide real differences in detecting actual language changes. Future work should include precision, recall, and F1-score specifically for the switch class.

8. Conclusions

Key Findings:

- GRU was the stronger baseline for Russian-English code-switch language modeling.
- CNN learned meaningful patterns, but underperformed in next-token prediction.
- Switch-point accuracy was similar across models.
- Recurrent memory appears useful for mixed-language streams.

Limitations:

- Synthetic dataset
- Small training subset
- Only compact architectures tested
- Short training schedule
- Accuracy-only switch metric